Development of the Respiratory Tract

Last updated March 22, 2026 13 min read ScholarRx

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Adapted from: Development of the Respiratory Tract

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Categorize the stages of lung development.
2. Outline the processes of branching, septation, alveolarization, and vascularization in lung development.

CASE CONNECTION

A 26-year-old patient who is pregnant presents at their 32-week mark with complications and has to deliver their baby early, before the 34-week mark. They come to you with concerns about their baby's ability to breathe after birth given that the baby is premature.

As you read this brick, think about how you would respond to the parent and address their concerns.

GO TO CONCLUSION

What Is the Development of the Respiratory Tract?

Pathology of the respiratory system in a newborn can result in serious illness, so it’s imperative to understand the timeline and process of its development, which continues for many years after birth.

Four main processes are essential to respiratory system development:

- Branching
- Septation (division of a single structure into two or more by formation divisions within the structure [septae])
- Alveolarization (formation of the thin alveoli, the gas exchange units of the lung)
- Vascularization (formation of pulmonary blood vessels to take blood to and from the alveoli)

Keep these important processes in mind as we walk through the stages of respiratory tract development, emphasizing lung development.

What Are the Five Stages of Lung Development? (LO#1)

The five stages of lung development are:

- Embryonic

- Pseudoglandular
- Canalicular
- Saccular
- Alveolar

Each is characterized by specific development stages. The stages overlap slightly because development is a fluid process, and cranial segments of the lung tend to mature faster than caudal segments.

Embryonic Stage (4–7 Weeks' Gestation)

All parts of the body develop from one of three layers: internal endoderm, external ectoderm, or mesoderm, which lies in the middle. The internal lining, or epithelium, of the airway develops from the endoderm. The cartilage and muscles of the respiratory airways and lung tissue itself all develop from the mesoderm.

The first sign of the respiratory system appears just 26 days after fertilization, when an increase in retinoic acid triggers formation of a **lung bud**, also called the respiratory diverticulum.

Development of the Lung Bud. The lung bud branches off the anterior aspect of the developing foregut, the part of the foregut that will later become the esophagus (Figure 1, left).

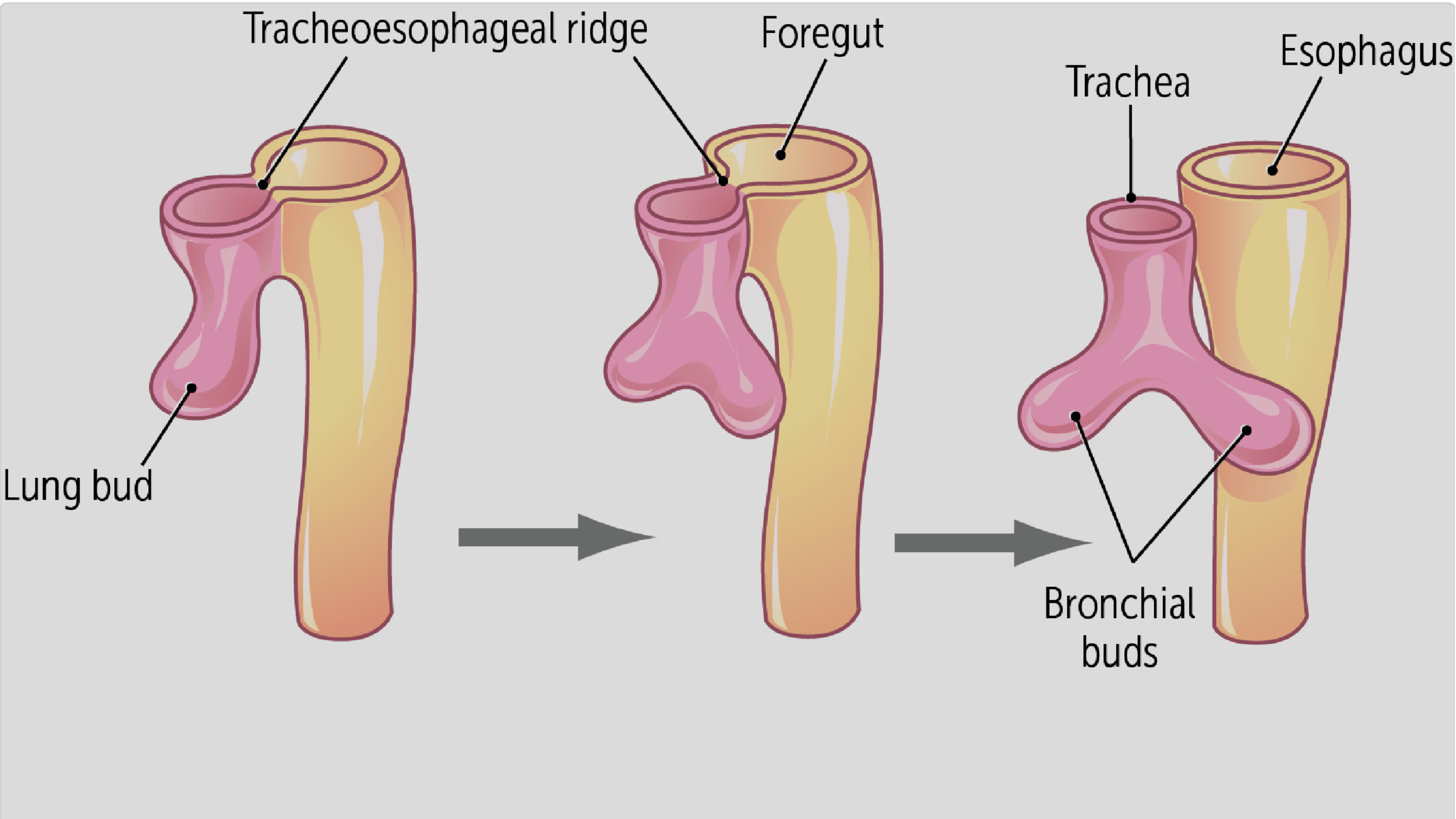


Figure 1

At first, the lung bud is open to the foregut. As it continues to develop, ingrowths on each side form **tracheoesophageal ridges** (see Figure 1). These ridges join in the midline to form the tracheoesophageal septum that separates two tubes: the trachea (ventral, or anterior) and the esophagus (dorsal, or posterior).

The newly formed trachea maintains its connection to the pharynx through the laryngeal orifice. A swelling in the upper airway tissue anterior to the orifice will later develop into the epiglottis, and the vocal cords will form around the orifice.

CLINICAL CORRELATION

The formation of the tracheoesophageal septum is vital, because abnormal septation can result in esophageal atresia (a blind pocket, i.e. the esophagus ends before getting to the stomach or does not connect to the stomach) or a tracheoesophageal fistula (tube that connects trachea and esophagus). These can lead to abnormal esophageal and respiratory function and severe pneumonia after birth.

The lung bud develops from the foregut.

Development of the Bronchial Buds. By 33 days of development, the single lung bud has divided into left and right bronchial buds that extend on either side of the esophagus. These buds will become the left and right main bronchi. Because of the position of the heart, the right main bronchus is more vertically oriented and is wider than the left main bronchus (Figure 2).

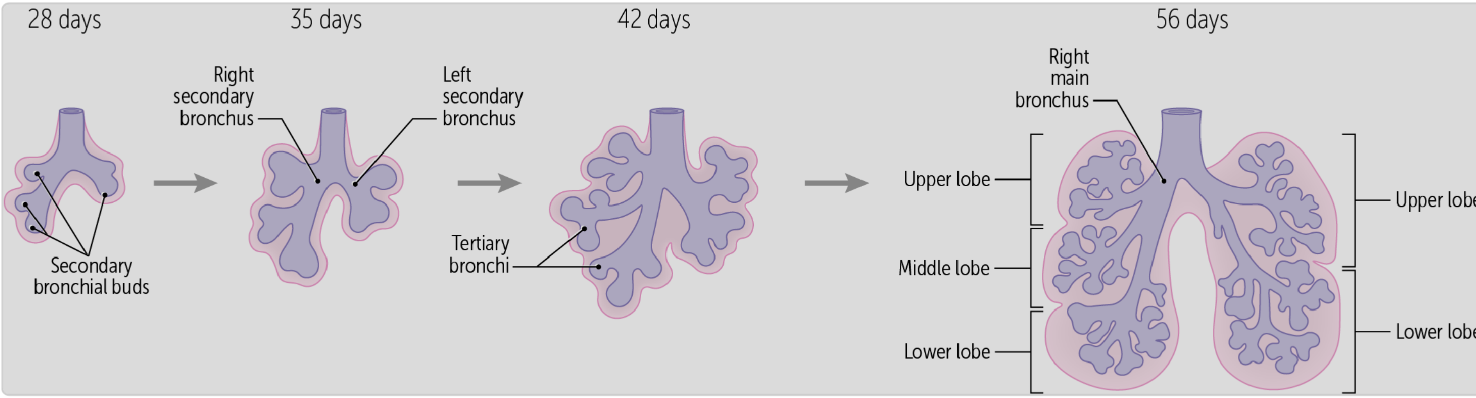


Figure 2

CLINICAL CORRELATION

Larger foreign bodies are almost always aspirated into the right main bronchus because it is wider and more vertically oriented than the left.

Further branching of the bronchial buds forms the secondary bronchi that will later branch to form the two lobar bronchi of the left lung and three lobar bronchi of the right lung.

Development of the Pulmonary Vasculature. As the bronchial tree branches, vascularization of the lungs has also begun. Mesoderm typically develops into the developing vascular and lymphatic systems.

Pulmonary arteries from the sixth aortic arch and pulmonary veins from the left atrium extend into the mesoderm surrounding the primitive airways.

Pseudoglandular Stage (5–17 Weeks' Gestation)

The pseudoglandular stage is marked by extensive branching of both airways and blood vessels, signaled by fibroblast growth factors (eg, FGF-10) secreted from surrounding mesoderm. The main bronchi branch first into bronchioles and then into terminal bronchioles, completing the conductive airway system.

A significant amount of mesoderm exists between the branching airways and forms vessels that branch near them, giving the primitive lungs a pseudoglandular appearance (Figure 3).

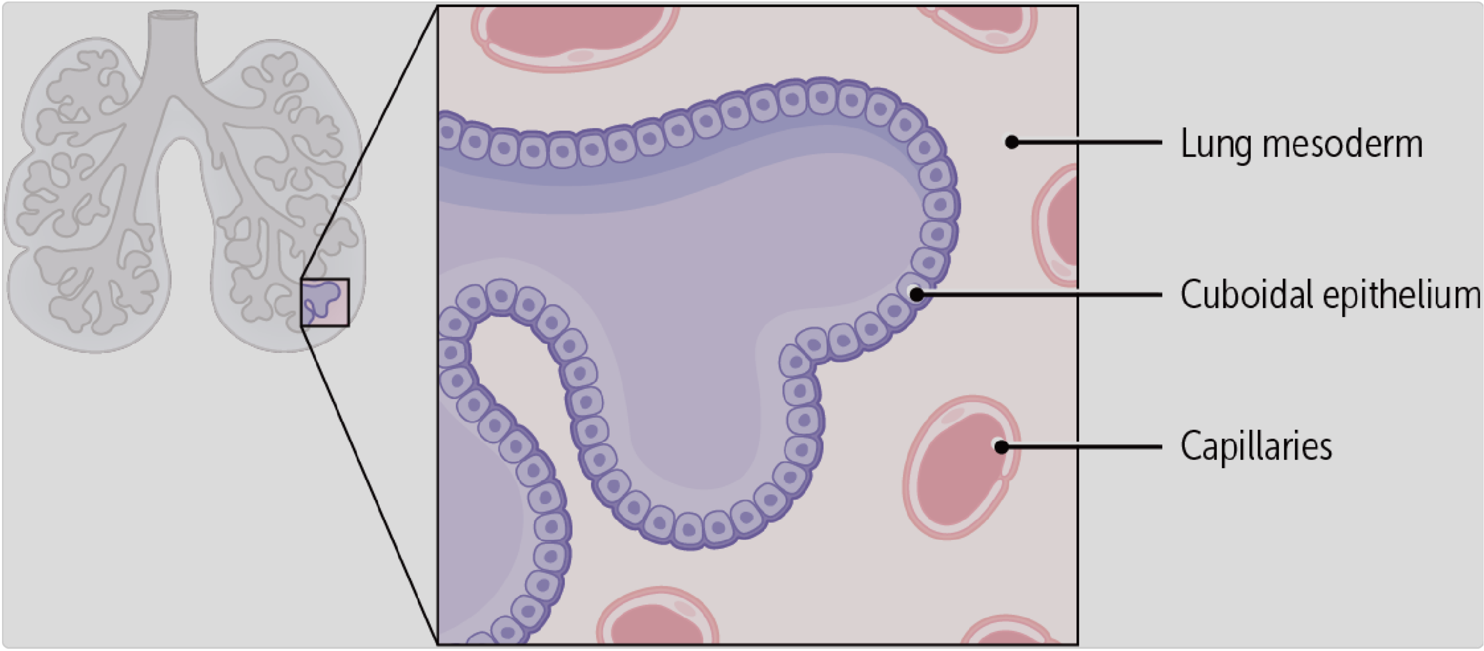


Figure 3

A transition from columnar to cuboidal epithelium occurs in these newer, distal airways. However, the cytoplasm of these cells is poorly differentiated, with few microvilli and very immature cilia. For this reason, and because respiratory bronchioles and alveoli are not yet present, the lungs cannot sustain life at this stage.

Canalicular Stage (16–25 Weeks' Gestation)

Additional branching occurs during the canalicular phase, by the end of which are 17 orders of branching. Respiratory bronchioles become prominent in this stage.

Rudimentary **acini** form during the canalicular stage. An acinus (a singular acini) consists of a respiratory bronchiole, alveolar ducts, and alveolar sacs; together, they comprise the nascent **respiratory zone** where gas exchange between the alveoli and pulmonary capillaries will eventually occur.

The primitive alveolar ducts form as an extension of the respiratory bronchioles. Some thin-walled terminal sacs (primordial alveoli) develop at the ends of the respiratory bronchioles. These can allow gas exchange to occur in an infant born toward the end of this stage (after 22 weeks' gestation), but the supporting structures are not yet fully in place to support adequate gas exchange within the developing lungs.

In this stage, peripheral growth of capillaries also increases significantly. They begin to distribute more uniformly, in closer contact with the newly established air spaces (Figure 4).

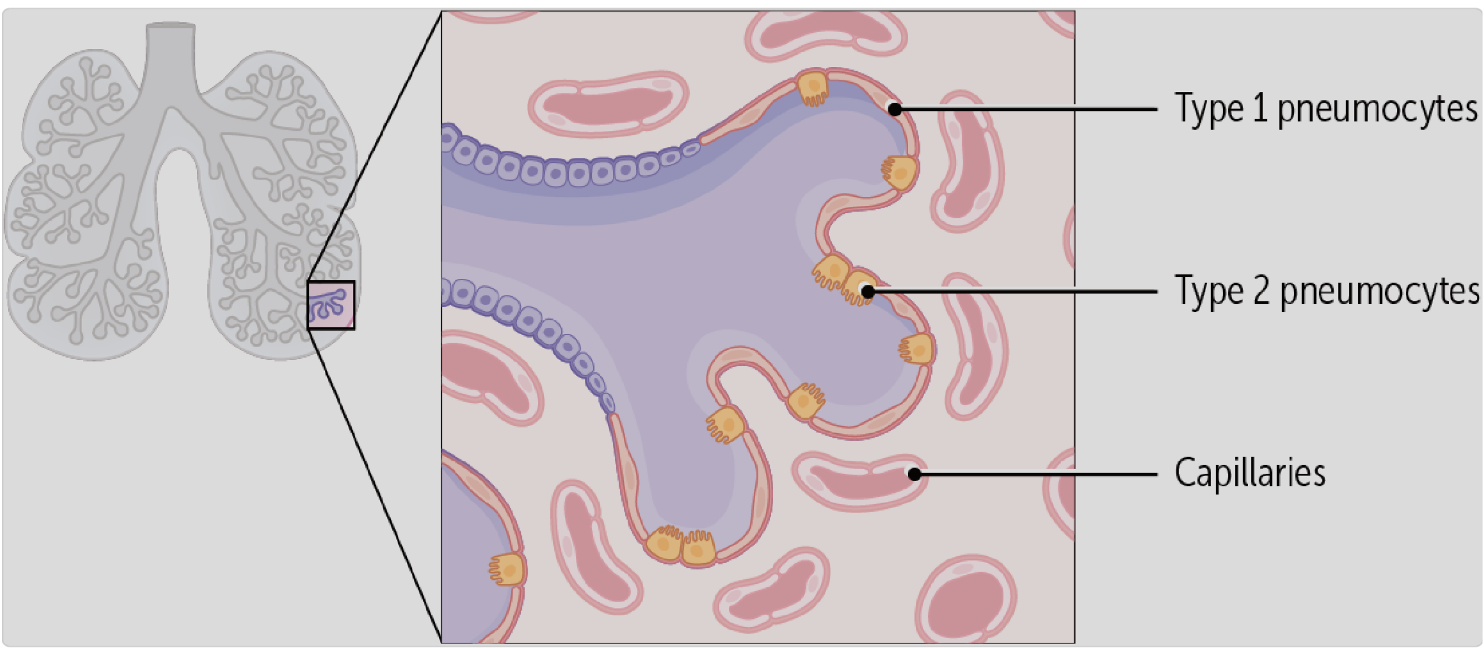


Figure 4

Saccular Stage (24 Weeks' Gestation to Birth)

During the saccular stage, many more terminal sacs (primordial alveoli) develop. Their epithelium becomes very thin. In addition, septation with elastic fibers begins to divide the alveolar sacs into smaller units. These have closer yet still incomplete contact with the capillaries required for gas exchange. Many capillaries begin to expand and press into the developing alveolar sacs.

Types 1 and 2 Pneumocytes. Within the acini, further development occurs in the type 1 and 2 pneumocytes that line the alveoli. Type 1 pneumocytes differentiate from the type 2 pneumocytes and become thinner to maintain contact with capillary endothelium, creating the blood-air barrier. This close contact allows for easy exchange of carbon dioxide and oxygen across adjacent cells.

Type 2 pneumocytes, on the other hand, create an air-alveolar interface by secreting **surfactant**, a phospholipid-rich fluid essential for lowering alveolar surface tension (Figure 5).

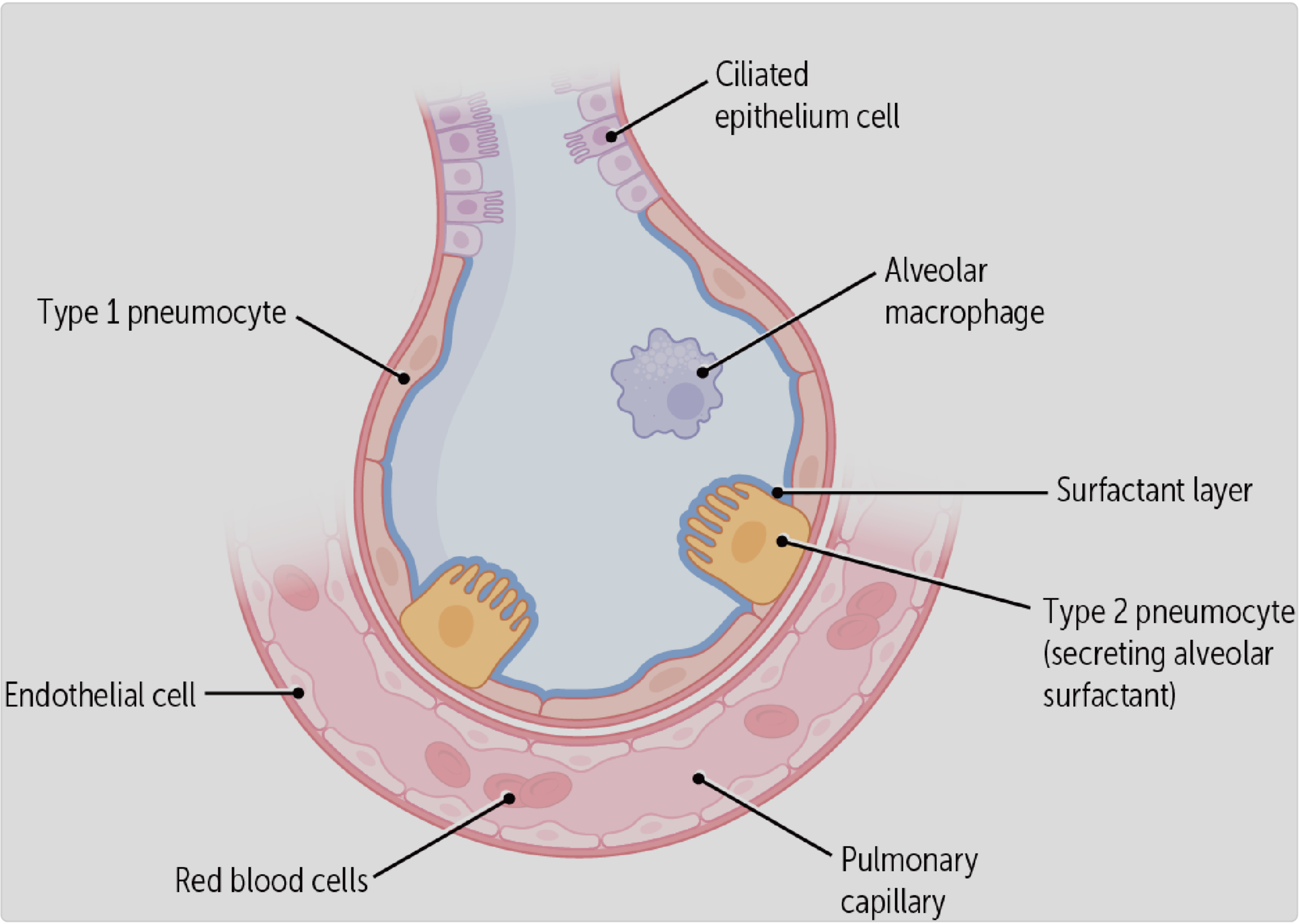


Figure 5

Production of surfactant begins at around 20 weeks but is initially present in very small amounts. Fetal viability depends on increased surfactant production that significantly increases after 22 weeks and then continues to increase throughout the term of the pregnancy.

Why is surfactant important? Alveolar sacs are elastic and subject to collapsing under the inward force of surface tension, which is caused by the adhesion of water molecules coating their surface. Surfactant, which is rich in lipids, decreases surface tension by interrupting the adhesive nature of these water molecules, so the alveolar sacs stay open. Without adequate surfactant, alveoli collapse during exhalation, limiting the surface area available for gas exchange.

CLINICAL CORRELATION

Physicians try to prevent premature labor until approximately 34 weeks, when physicians think there is enough surfactant for an infant to breathe on their own. In fact, when a patient goes into labor before 34 weeks, obstetricians give corticosteroid medications to increase surfactant production in the fetal lungs before delivery. If enough surfactant is not present at birth, an infant may develop neonatal respiratory distress syndrome.

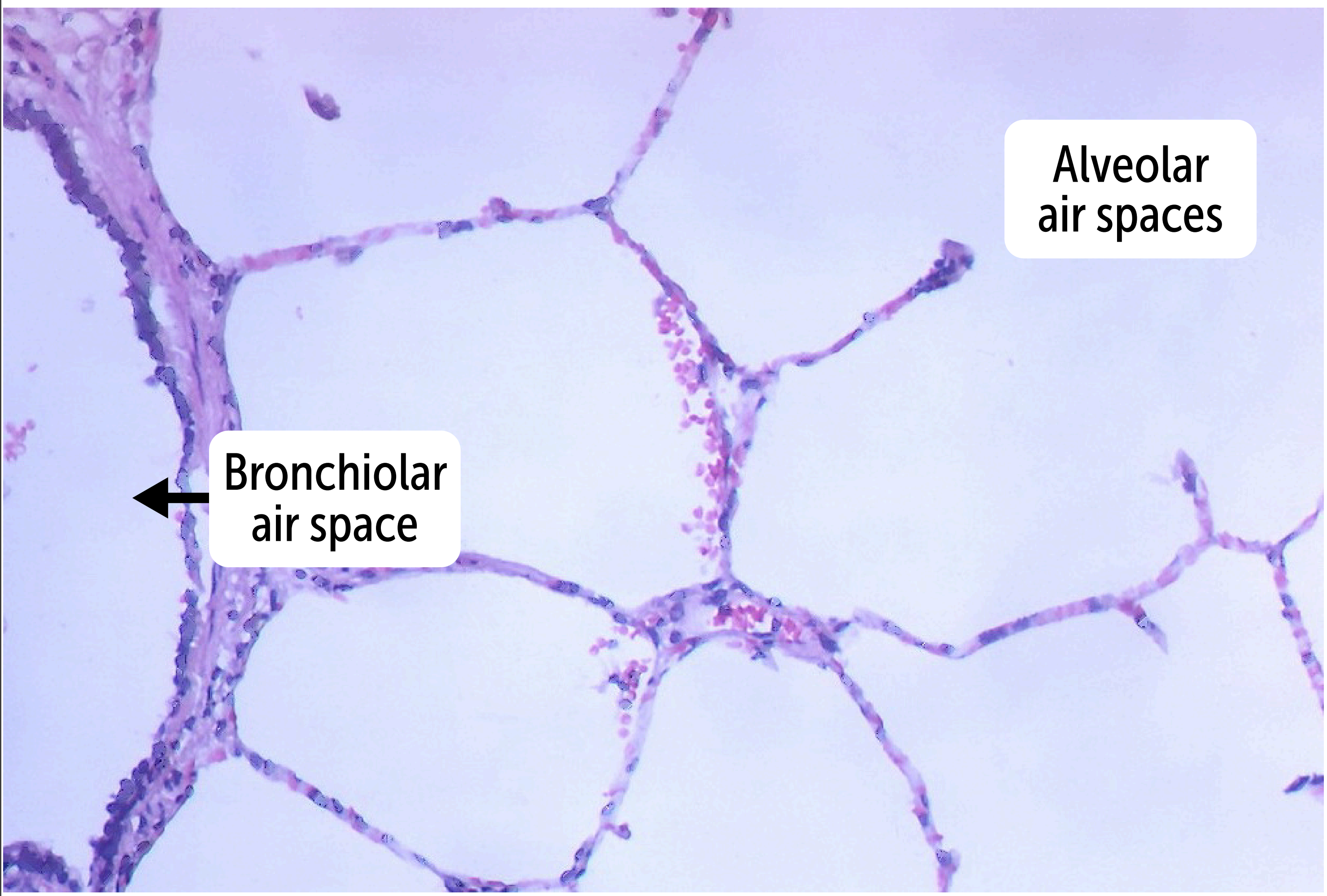
There is robust evidence that a single course of antenatal corticosteroids (i.e., 24 mg of betamethasone or dexamethasone), when there is a risk of preterm birth (at less than 34 weeks of gestation), reduces the risk of child death, regardless of resource level.

Type 2 pneumocytes produce surfactant.

Alveolar Stage (32 Weeks' Gestation to Age 8 Years)

Alveolarization occurs during this stage, as the internal surface area of the lung increases via an increase in acini, along with greater vascularization of the alveolar sacs.

Despite this increased alveolarization, at birth, only about 5% of the alveoli are mature; the remaining 95% mature after birth. Lung maturation continues until age 8 years, when 300 million mature alveoli have developed (Figure 6).



QUIZ

Tap image for quiz

Figure 6

Breathing Movements and the Initial Breath

Inspiration-like movements begin early in utero (as early as week 10), leading to aspiration of amniotic fluid into the developing lungs. This action provides a stimulating factor for lung development and early training of respiratory muscles. During late pregnancy, however, inspiration-like movements stop as the fetus prepares for birth. Important note, there is no actual breathing happening, just practice.

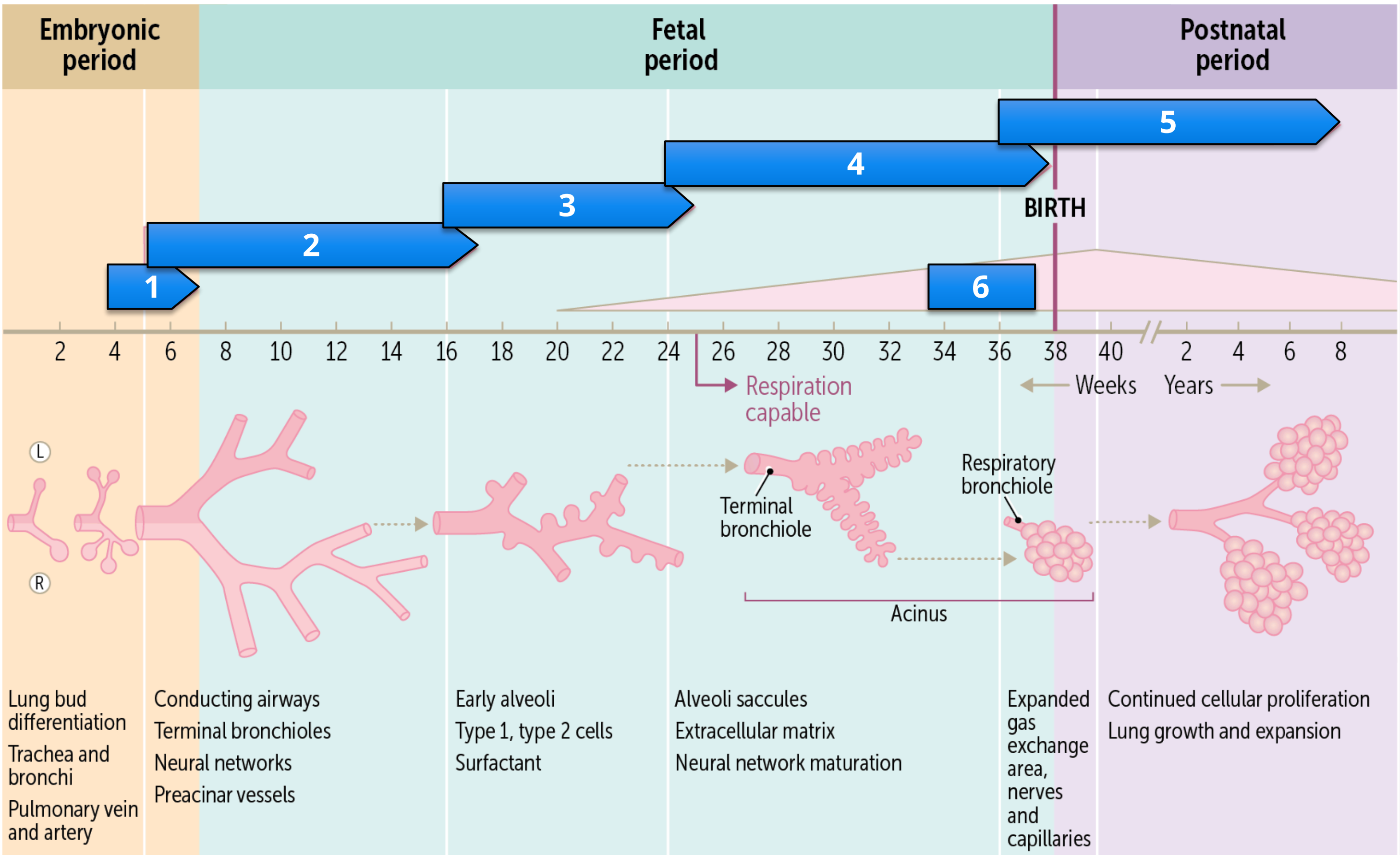
At birth, the first breath triggers rapid resorption of pulmonary (amniotic) fluid into alveolar capillaries and the local lymphatic system. Some of this fluid is also lost through the mouth and nose because of the pressure exerted on the fetal thorax during a vaginal delivery.

When air finally moves into the alveolar sacs, the lungs rapidly expand and fill the pleural cavity, maintaining lung expansion on exhalation. The initial breath is incredibly important for establishing internal pressure for multiple organ systems. This is often why physicians used to spank the child to get it to cry after birth to ensure they took a big first breath.

The amniotic fluid exits the airways through the mouth and nose because of the pressure on the thorax during delivery, and some of it is resorbed into alveolar capillaries and local lymphatic system.

Test your knowledge of the various stages of lung development below:

Directions: Drag and drop the terms below to the numbered areas where they best fit.



Saccular | Alveolar | Canalicular | Surfactant | Pseudoglandular | Embryonic

Submit

What Else Is Occurring During Lung Development? (LO#2)

Let's circle back to the four main processes mentioned at the beginning of the brick. As you have gathered, the processes of branching, septation, alveolarization, and vascularization are all occurring at different time points during lung development. In many cases they either span across multiple stages or even continue post birth. The following is a brief explanation of each of those four processes. Please note, the timelines are a general guideline. Things can occur a little earlier or later than noted below.

Branching: This process begins in the embryonic stage, around 26 days' gestation, when a lung bud forms from the foregut. Extensive airway branching occurs during the pseudo glandular stage, which spans from approximately 5 to 17 weeks of gestation. This stage completes the development of the conductive airway system from the trachea through the terminal bronchioles.

Septation: Septation involves the formation of septa, which are walls or partitions that divide the lung into distinct lobes and segments. This process is crucial for creating the separate compartments within the lungs that will house the alveoli. Septation occurs primarily during the canalicular stage, which takes place from around 16 to 25 weeks of gestation.

Alveolarization: This process continues long after birth during the alveolar phase, which starts around 36 weeks of gestation and extends into early childhood (around 8 years old). Alveolarization also involves the maturation of type 1 and type 2 pneumocytes, which are essential for gas exchange and surfactant production. This occurs during the saccular stage, from approximately 24 weeks of gestation to birth.

Vascularization: Vascularization involves the development of the blood vessels within the lungs, ensuring that the alveoli are well-supplied with blood for efficient gas exchange. The formation of the pulmonary vasculature begins early in lung development and continues to mature throughout the fetal period and after birth. This process is ongoing and critical for the proper function of the respiratory system.

CASE CONNECTION

BACK TO INTRODUCTION

You speak to the patient and hear their concerns. You inform them that you will administer a corticosteroid injection (typically betamethasone) to the pregnant patient, commonly called the lung maturation shot, to accelerate the development of the alveoli a few days before the baby is born. This will allow the baby to develop the alveoli needed to breath on their own and reduces the risk of lung problems.

Summary

- Four main processes are essential to respiratory system development: branching, septation, alveolarization, and vascularization.
- Lung development begins in the embryonic stage when a lung bud forms from the foregut at 26 days' gestation, triggered by retinoic acid signaling.
- Development of the conductive airway system (trachea through terminal bronchioles) is completed after extensive airway branching occurs during the pseudoglandular stage.
- The canalicular stage marks the beginning of true respiratory airway development, allowing for a potentially viable fetus beyond 22 weeks' gestation.
- Maturation of type 1 and type 2 pneumocytes occurs during the saccular stage, which allows alveoli to function in gas exchange, produce surfactant, and maintain patency.
- Lung maturation continues long after birth during the alveolar phase.

Review Questions

1. Which of the following is the principal purpose of alveolar surfactant?

A. Decrease alveolar surface area

B. Decrease alveolar surface tension

C. Increase alveolar surface area

D. Increase alveolar surface tension

E. Maintain pleural and visceral layer contact

Hide Explanation

The correct answer is decrease alveolar surface tension (B). Surfactant is a phospholipid-rich substance secreted by type 2 pneumocytes. It coats alveoli and decreases surface tension at the air-liquid interface in the lungs, preventing the collapse of alveoli and small airways during expiration. Surfactant has no effect on the surface area of the lung; the sheer number of alveoli and their internal septations determine the surface area available for gas exchange in the lung, so decrease alveolar surface area (A) and increase alveolar surface area (C) are incorrect. Increased alveolar surface tension (D) would cause the alveoli to collapse. Maintain pleural and visceral layer contact (E) is incorrect because the pleural and visceral layers of the lung are separated by a very thin layer of pleural fluid, not surfactant.

2. A 28-year-old patient who is pregnant is at 20 weeks of gestation. An ultrasound reveals normal lung development in the fetus. Which of the following processes is primarily occurring in the fetal lungs at this stage of gestation?

A. Branching

B. Septation

C. Alveolarization

D. Vascularization

Hide Explanation

The correct answer is septation (B). Septation involves the formation of septa, which divide the lung into distinct lobes and segments. This process occurs primarily during the canalicular stage, which takes place from around 16 to 25 weeks of gestation. Therefore, at 20 weeks of gestation, septation is the primary process occurring in the fetal lungs. Branching (A) begins in the embryonic stage around 26 days' gestation and continues extensively during the pseudoglandular stage, which spans from approximately 5 to 17 weeks of gestation. By 20 weeks, the branching process is largely complete. Alveolarization (C) starts around 36 weeks of gestation and extends into early childhood (around 8 years old). This process involves the formation and maturation of alveoli, the tiny air sacs in the lungs where gas exchange occurs. It is not the primary process at 20 weeks of gestation. Vascularization (D) involves the development of blood vessels within the lungs and continues to mature throughout the fetal period and after birth. While it is an ongoing process, it is not the primary focus at 20 weeks of gestation.

3. What is the primary role of Type 2 pneumocytes in lung development?

A. Bronchial tree formation

B. Surfactant production

C. Lung vascularization

D. Gas exchange

E. Trachea formation

Hide Explanation

The correct answer is surfactant production (B). Type 2 pneumocytes are responsible for the production of surfactant, a phospholipid-rich fluid that is essential for lowering alveolar surface tension and preventing the collapse of alveoli during exhalation. They do not contribute to the formation of the bronchial tree (A), which is formed through the process of branching. They also do not directly contribute to the vascularization of the lungs (C), which is primarily the responsibility of the mesoderm. While they do contribute to gas exchange (D), this is not their primary role. Finally, they do not contribute to the formation of the trachea (E), which is formed from the foregut.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ~~Categorize the stages of lung development.~~

02 ~~Outline the processes of branching, septation, alveolarization, and vascularization in lung development.~~

Objective Confidence: 2/2